Google Play Store Apps – Project Explanation

Project type: Exploratory Data Analysis (EDA) | **Tools:** Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

Project Overview

This project explores Google Play Store app data to understand ratings, installations, pricing, reviews, categories, and update activity. The goal is to turn raw records into useful insights about app performance and user preferences.

Requirements

The analysis investigates which categories have the most apps and installs, how Free and Paid apps compare, which categories receive higher ratings, whether reviews relate to installs, and how app update activity changes over time.

Data Preparation & Approach

The notebook removes a corrupted category record and duplicate app names, converts installs and prices to numeric values, standardizes text fields, converts update dates, and checks rating values. It then performs basic, intermediate, and advanced analysis using summaries, comparisons, and visualizations.

Key Findings

Dataset	9,658 apps across 33 categories
Average rating	4.17
App type	8,902 Free apps; 756 Paid apps
Highest installs	Games, followed by Communication and Tools
Highest average category rating	Events (approximately 4.44)
Rating threshold	6,286 apps rated 4.0 or higher
Reviews and installs	Moderate positive correlation (approximately 0.63)

Challenges

The main challenges were inconsistent formats in installs and prices, duplicate records, missing values, and selecting visualizations that make comparisons easy to interpret.

Recommendations

Evaluate apps using ratings and reviews alongside installs; study high-demand categories while considering competition; and use user feedback to guide updates. Future work could explore pricing, review-to-install ratios, and app-level performance.

Outcome

The project strengthened practical skills in Python data cleaning, exploratory analysis, and visualization, and demonstrated how app data can support informed product decisions.

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